Prevalence and Determinants of Postpartum Depression among Adolescent and Adult Mothers in Khulna region, Bangladesh

Course Title: Masters Level Seminar

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Background

- Postpartum Depression (PPD) prevalence: Globally 17.22%; LMICs 18.6%; Bangladesh 27.75%
- Globally and in LMICs, adolescent mothers have higher PPD prevalence than adults; however, in Bangladesh adult mothers have higher PPD prevalence.
- 30 risk factors and 5 protective factors for PPD were identified among Bangladeshi postpartum mothers.

Justifications of the Study

- No comparative cross-sectional study was conducted to estimate and compare adolescent and adult PPD prevalence in Bangladesh
- Risk factors and protective factors may vary among adolescent and adult postpartum mothers
- Breastfeeding practice and reproductive seasons were not explored in earlier studies which may act as protective factors

Research Questions

Primary Research Question

- What are the prevalence of PPD among among adolescent and adult mothers in Khulna region, Bangladesh?

Secondary Research Questions

- Is PPD associated with self-esteem among adolescent and adult mothers in Khulna region, Bangladesh, after adjusting for socio-demographic covariates?
- Are breastfeeding practice and reproductive seasons associated protectively with PPD among adolescent and adult mothers in Khulna region, Bangladesh?

Objectives

General Objective

- To assess the prevalence and determinants of PPD among adolescent and adult mothers in Khulna region, Bangladesh

Specific Objectives

- To estimate the prevalence of PPD among adolescent and adult mothers
- To measure the association between participants self-esteem and PPD after adjusting for socio-demographic covariates
- To measure the association between breastfeeding practice and PPD; between reproductive seasons and PPD, after adjusting for socio-demographic covariates

Conceptual Framework

Independent Variables

- Socio-demographic Factors
  - Maternal Age
  - Education Level
  - Maternal Occupation
  - Household Income
  - Religion
  - Residence
- Self-esteem
- Reproductive seasons
- Breastfeeding practice
- Newborn gender
- Parity

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Dependent Variable

Postpartum Depression (EPDS Score $\geq$10)

Research Methodology

- **Study Design:** Comparative cross-sectional study
- **Study Population:** Adolescent and adult postpartum mothers in Khulna region
- **Study Site:** Khulna Medical College and Gazi Medical College
- **Study Duration:** Between July 2026 and September 2026

Research Methodology

- **Inclusion Criteria:**
  - Adolescent mothers (age 14-19 years) within 1 year postpartum
  - Adult mothers (age 20-49 years) within 1 year postpartum
- **Exclusion Criteria:**
  - Mothers having newborns with severe illness
  - Mothers having chronic illness
  - Mothers having preexisting mental health disorder

Sample Size Calculation

$$n = \frac{r+1}{r} \cdot \frac{\bar{p}(1-\bar{p})(Z_\beta + Z_{\alpha/2})^2}{(p_1 - p_2)^2}$$
$$= \frac{(1+1)}{1} \cdot \frac{0.33(1-0.33)(1.96+0.84)^2}{(0.26-0.40)^2}$$
$$= 176 \text{ per group}$$

Assumptions

- $p_1 = 0.26$ (Teen)
- $p_2 = 0.40$ (Adult)
- $\bar{p} = \frac{0.26+0.40}{2} = 0.33$
- $Z_{\alpha/2} = 1.96$
- $Z_\beta = 0.84$
- $r = 1$

$$\text{Total} = (176 + 176)$$
$$= \mathbf{352}$$

Thank You